

204. Count Primes

Solved

Medium Topics Companies Hint

Given an integer n , return the number of prime numbers that are strictly less than n .

Example 1:

Input: $n = 10$

Output: 4

Explanation: There are 4 prime numbers less than 10, they are 2, 3, 5, 7.

Example 2:

Input: $n = 0$

Output: 0

Example 3:

Input: $n = 1$

Output: 0

8.7K 136 67 Online

Java Auto

```

1 class Solution {
2     public int countPrimes(int n) {
3
4     }
5 }

```

 $n = 10$
 $1, 2, 3, 4, 5, \dots \rightarrow 4 \text{ prime}$
 $6, 7, 8, 9, 10 \rightarrow \text{not}$

```

for(int i=1; i<=10; i++) {
    isPrime(i); // count++
}

```

Saved

Talking: Coding Blocks

Test Result

Testcase

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DSA-2.0 Day 22 src Solution

CountPrimes.java

```

class Solution {
    public int countPrimes(int n) {
        int count = 0;
        for(int i=1; i<n; i++) {
            if(isPrime(i)) {
                count++;
            }
        }
        return count;
    }
}

1 usage
public boolean isPrime(int x) {
    if(x == 1) {
        return false;
    }
    for(int d=2; d<=x; d++) {
        if(x%d == 0) {
            return false;
        }
    }
    return true;
}

```

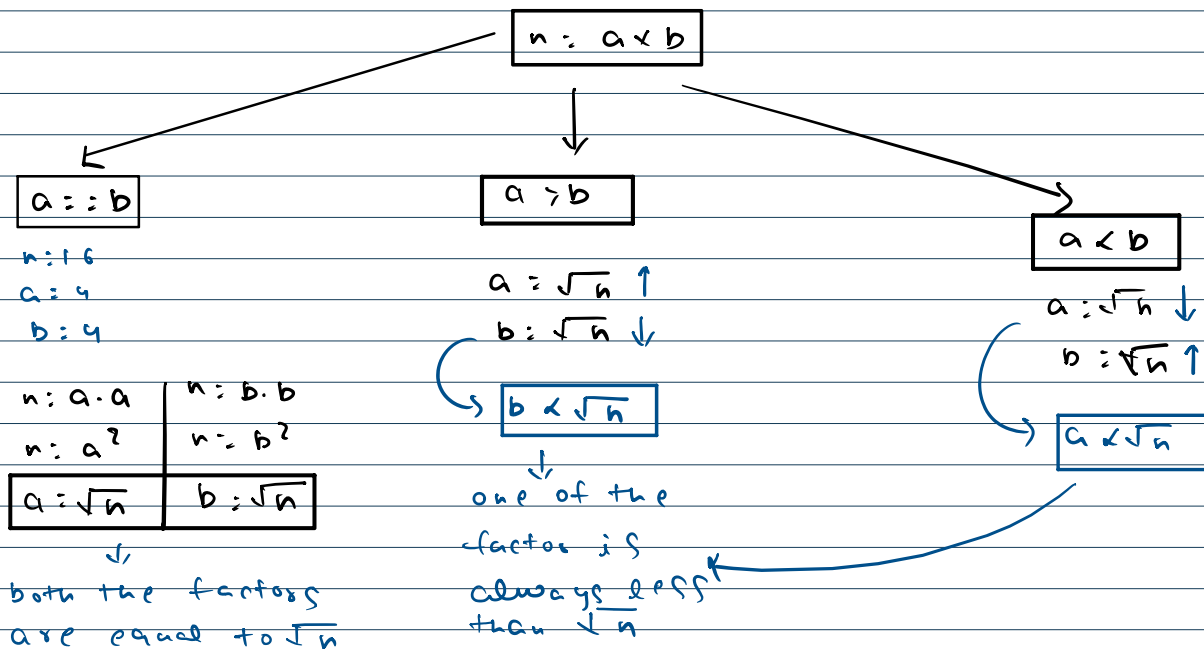
Time $\Rightarrow n$: Input dependent
 $\rightarrow O(n)$
 $\Downarrow$
 $O(n) \times O(n) \Rightarrow O(n^2)$
 $O(n \cdot n)$
 $\Downarrow$
 $O(n^2)$
 $\rightarrow O(x)$
 $\text{constant} \Rightarrow n = 10^6$
 $O(n^2) = (10^6)^2 = 10^{12}$
 insts

TLE


 $\rightarrow \text{Instructions}$
 $\downarrow$
 10^8 inst.
 $n \rightarrow \text{numbers } \geq 1, > 0 \Rightarrow \text{natural number}$
 $n \begin{cases} \rightarrow \text{Prime} \\ \rightarrow \text{not prime} \Rightarrow \checkmark \end{cases}$

 number is not prime $\Rightarrow$ it has factors other than 1.
 (n)

$$n = a \times b \quad a, b \neq 1$$



```

class Solution {
public:
    int countPrimes(int n) {
        int count = 0;
        for(int i=1; i<n; i++) {
            if(isPrime(i)) {
                count++;
            }
        }
        return count;
    }

    2 usages
    public:
    boolean isPrime(int x) {
        if(x == 1) {
            return false;
        }
        for(int d=2; d*d<=x; d++) {
            if(x%d == 0) {
                return false;
            }
        }
        return true;
    }
}
  
```

Time: $O(n \cdot \sqrt{n})$

$O(n \cdot n^{1/2})$

$n^{1 + \frac{1}{2}}$

$n^{\frac{2+1}{2}} = n^{\frac{3}{2}}$

$n = 10^6$
 $10^6 \times \frac{3}{2} = 10^9$
 10⁹ is not acceptable.

>
 ↓
 TLE



10⁹
 is not acceptable

Sieve of Eratosthenes

find all prime numbers upto n .

2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19
20	21	22	23	24	25	26	27	28

$$n = 120 \Rightarrow \frac{n}{2} + \frac{n}{3} + \frac{n}{5} + \frac{n}{7} + \frac{n}{11}$$

11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100
101	102	103	104	105	106	107	108	109	110
111	112	113	114	115	116	117	118	119	120

$$\sqrt{n} = \sqrt{120} \approx 11 \text{ (int)}$$

{ 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37,
 41, 43, 47, 53, 59, 61, 71, 73, 83,
 89, 97, 101, 103, 107, 109, 113 }

prime numbers

$$n = 1024 = 2^{10}$$

$$\frac{n}{2} + \frac{n}{3} + \frac{n}{5} + \frac{n}{7} + \frac{n}{11} + \dots$$

$$\log_2 \log_2 N$$

$$\log_2 \log_2 2^{10}$$

$$\log_2 10 (\log_2 2)$$

$$\log_2 10 \cdot 1$$

$$\hookrightarrow \text{approx.} = 3$$

$$\downarrow$$

 count cut

$$\downarrow$$

$$O(1)$$

series length

$$\downarrow$$

$$N$$

$$n \left[\frac{1}{2} + \frac{1}{3} + \frac{1}{5} + \frac{1}{7} + \frac{1}{11} + \dots \right]$$

$$n \times \lceil \log_2(\log_2 N) \rceil$$

$$O(n) \cdot O(1) \sim O(n)$$

Merge 2 Sorted Arrays

arr1 →

2	3	5
---	---	---

 (n)

arr2 →

1	3	6	7	8
---	---	---	---	---

 (m)

arr (n+m)

1	2	3	3	5	6	7	8
---	---	---	---	---	---	---	---

arr1 →

2	3	5
---	---	---

 (n)

arr2 →

1	3	6	7	8
---	---	---	---	---

 (m)

